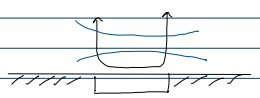
Lecture 22

Tuesday, 4 April 2023 1:29 PM

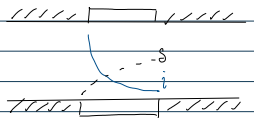
Impact of electrode Conductivity on Current dist.

So far, we have shown current non-uniformities arise from: • electric field / geometry (primary, secondary)

We can also encounter current variations due to imperfect conductors (i.e. resistive substrates)

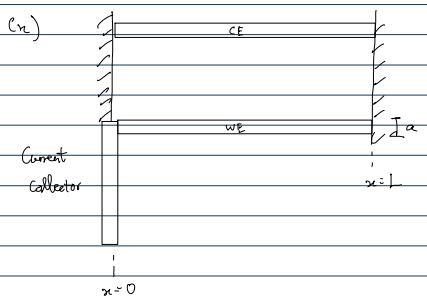


• MT effects during fast kinetics



Current dist. in Resistive Substrate:

↳ Encountered frequently during metallization of Si chips



* previously we set $V=\phi$ @ $y=0$ (1D problem w/ uniform current dist.)

potential of metal

Now because substrate is resistive: all current must flow through $x=0$, $\phi_m = f(x)$

$\eta = \phi_m - \phi_s$ (potential of metal; potential in solution)

To solve, we assume $\nabla c_j = 0$, then

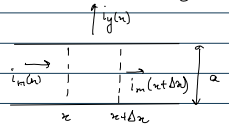
$$\nabla^2 \phi_s = 0$$

$$\frac{\partial^2 \phi_s}{\partial x^2} + \frac{\partial^2 \phi_s}{\partial y^2} = 0$$

, for BC's: Assume Tafel kinetics @ electrode

$$i(x) = -\kappa \left.\frac{\partial \phi_s}{\partial y}\right|_{y=0} = -i_o \exp\left[\frac{-\alpha_a F}{RT}(\phi_m - \phi_s)\right]$$

$= f(x)$ → We get this from charge balance



In − out + gen = Acc.

$$a\left[i_m(x) - i_m(x+\Delta x)\right] - i_y(x)\Delta x = 0$$

$$i_y(x) = \frac{(i_m|_x - i_m|_{x+\Delta x})\,a}{\Delta x}$$

$$i_y(x) = -a\frac{di_m}{dx}$$

, From Ohm's law: $i_y(x) = \frac{d}{dx}\left(\frac{1}{R_s}\frac{d\phi_m}{dx}\right)$

$i_y(x) = -a \frac{di_m}{dx}$; From Ohm's law: $i_y(x) = \frac{d}{dx}\left(\frac{1}{R_s}\frac{d\phi_m}{dx}\right)$

R_s: Sheet resistance

Combining:

$$-a\frac{di_m}{dx} = \frac{d}{dx}\left(\frac{1}{R_s}\frac{d\phi_m}{dx}\right)$$

$$\boxed{i_m = \frac{-1}{aR_s}\frac{d\phi_m}{dx} = -\sigma\frac{d\phi_m}{dx}}$$

σ: analogous to κ [$\Omega^{-1}\,cm^{-1}$]

gradient is zero perpendicular to the insulator position

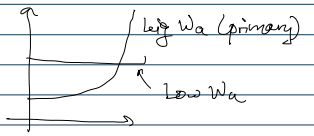
; BC's: @ $x = L$, $\frac{d\phi_m}{dx} = 0$

@ $x = 0$, $\phi_m = V$

Can be solved numerically or analytically (limiting cases):

+ Depends on $Wa_1 = \frac{RTK}{\alpha_c F i_{lim} L}$; $K_0 = \frac{KL}{a\sigma}$ (ratio of electrolyte to substrate conc.)

If $\frac{Wa}{K_0}$ → large ; } more uniform



$K_0 \downarrow$, $\sigma \uparrow$ (electrode cond. goes up) → tend to be more uniform

Summary:

1) Spatial variations of current on electrode are caused by:

→ $Wa = \frac{R_{ct}}{R_\Omega} \to 0$ when, large

$i_{avg} << i_{lim}$

→ MT conditions when $i_{avg} \approx i_{lim}$

→ Resistive electrodes (even when $Wa > 0.1$)

Uniform current:

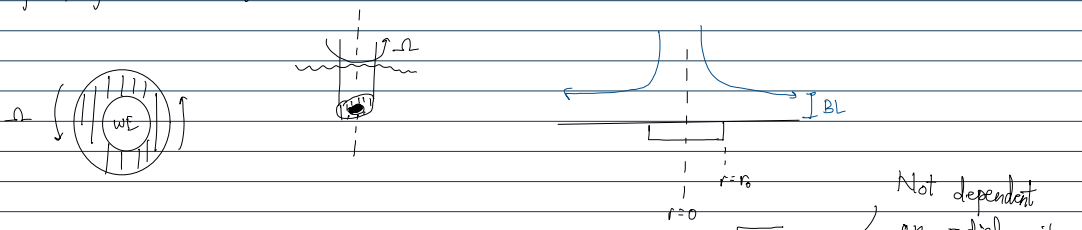
1) Conductive electrolyte
2) Small electrodes
3) Uniform MT if $i \sim i_{lim}$
4) Conductive electrodes

Fluid Flow to a rotating disk (RDE)

→ Even under i_{lim} we have non-uniformity of current due to MT (Boundary layer problems)

→ A rotating disk electrode provides a MT scenario where an exact solution to fluid flow is obtained

→ A rotating disk electrode provides a MT scenario where an exact solution to fluid flow is obtained



↳ Solution to flow flow reveals a momentum BL of thickness: $\sqrt{\frac{\nu}{\Omega}}$ (Thicker than diffusion BL) ← Not dependent on radial position

↳ Based on this thickness, we can define dimensionless axial variable

$$\xi = \frac{z}{\sqrt{\frac{\nu}{\Omega}}}$$

Then we can define each velocity component in non-dim form as func. of ξ

Radial · $v_r = r\Omega f(\xi)$

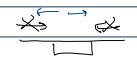
Angular · $v_\theta = r\Omega G(\xi)$

Axial · $v_z = \sqrt{\nu\Omega}\, H(\xi)$; No slip condition @ $\xi=0 \Rightarrow F=H=0$; $G=1$ → whatever is at surface keeps rotating w/ it

↳ Axial velocity is not a function of radial position (Uniform diffusion layer thickness) → uniform current @ i_{lim}

↳ Radial velocity is outward (positive r) → negligible flux from large radial positions to smaller radial positions → minimal edge effects

} These enable its use (v. diff. from earlier case)



Now solving for current distribution:

↳ Assume limiting current @ disk, and $t_i = 0$ } Convection-diffusion equation

$$v_z\left(\frac{\partial c}{\partial z}\right) = D\left(\frac{\partial^2 c}{\partial z^2} + \frac{\partial^2 c}{\partial r^2}\right)$$

0 Diffusion layer constant in r direction (even though we have flow in r-direction) → No conc. gradients in the r-direction

From hydrodynamic analysis:

$$v_z = -a z^2 \left(\frac{\Omega}{\nu^{1/2}}\right)^{3/2}$$

* This is the 1 term Taylor expansion of $v_z = \sqrt{\nu\Omega}\cdot H(\xi)$

a: 0.51023 (fit parameter)

↓

Non-dimensionalize: $\bar{z} = \left(\frac{\Omega}{\nu^{1/6} D^{1/3}}\right)^{1/2} z$,

$$\therefore \frac{\partial c}{\partial z} = \frac{\partial c}{\partial \bar{z}}\left(\frac{\Omega}{\nu^{1/6} D^{1/2}}\right)^{1/2} \quad \frac{\partial^2 c}{\partial z^2} = \frac{\partial^2 c}{\partial \bar{z}^2}\left(\frac{\Omega}{\nu^{1/3} D^{2/3}}\right)$$

Plugging in:

$$-a z^2 \frac{dc}{d\bar{z}} = \frac{d^2 c}{d\bar{z}^2}$$

BC's: @ $\bar{z}=0$, $c=0$

Plugging in:

$$-a\bar{z}^2\frac{dc}{d\bar{z}} = \frac{d^2c}{d\bar{z}^2}$$

BC's: @ $\bar{z}=0$, $c=0$

@ $\bar{z}=\infty$, $c=c_\infty$